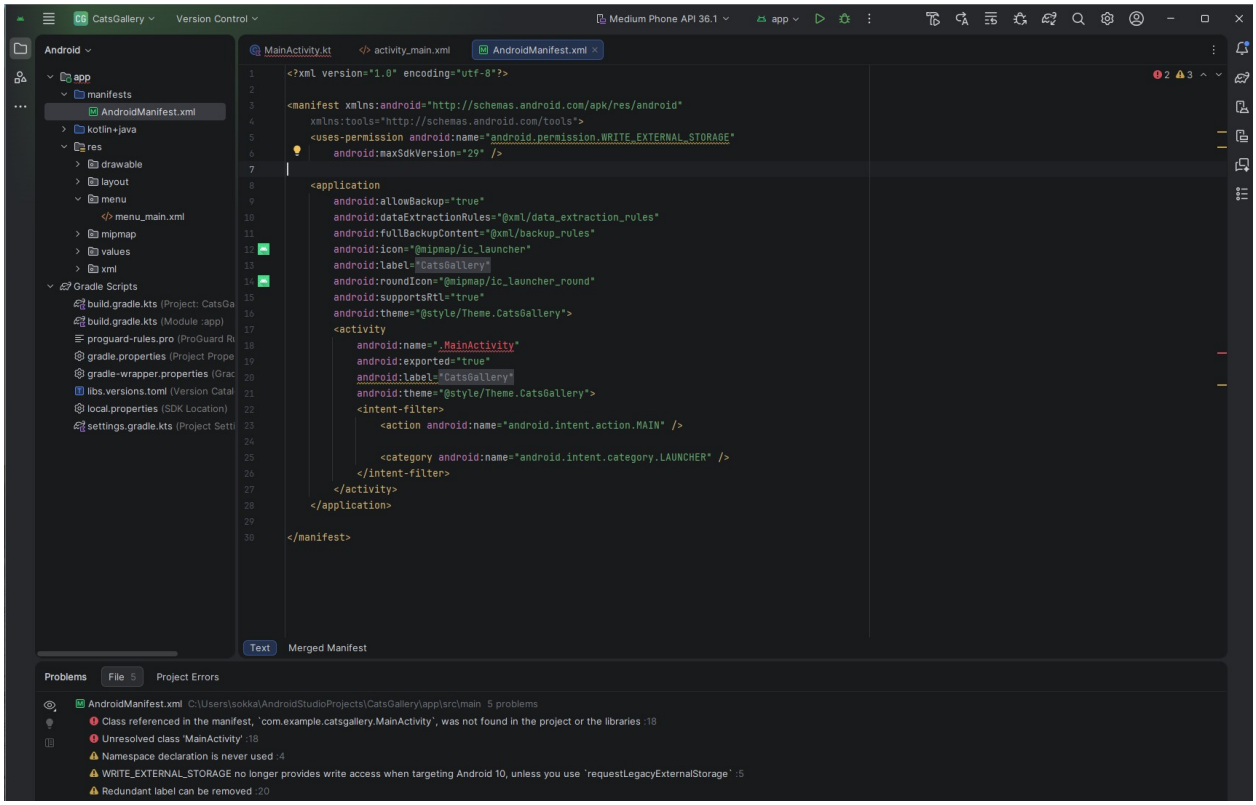
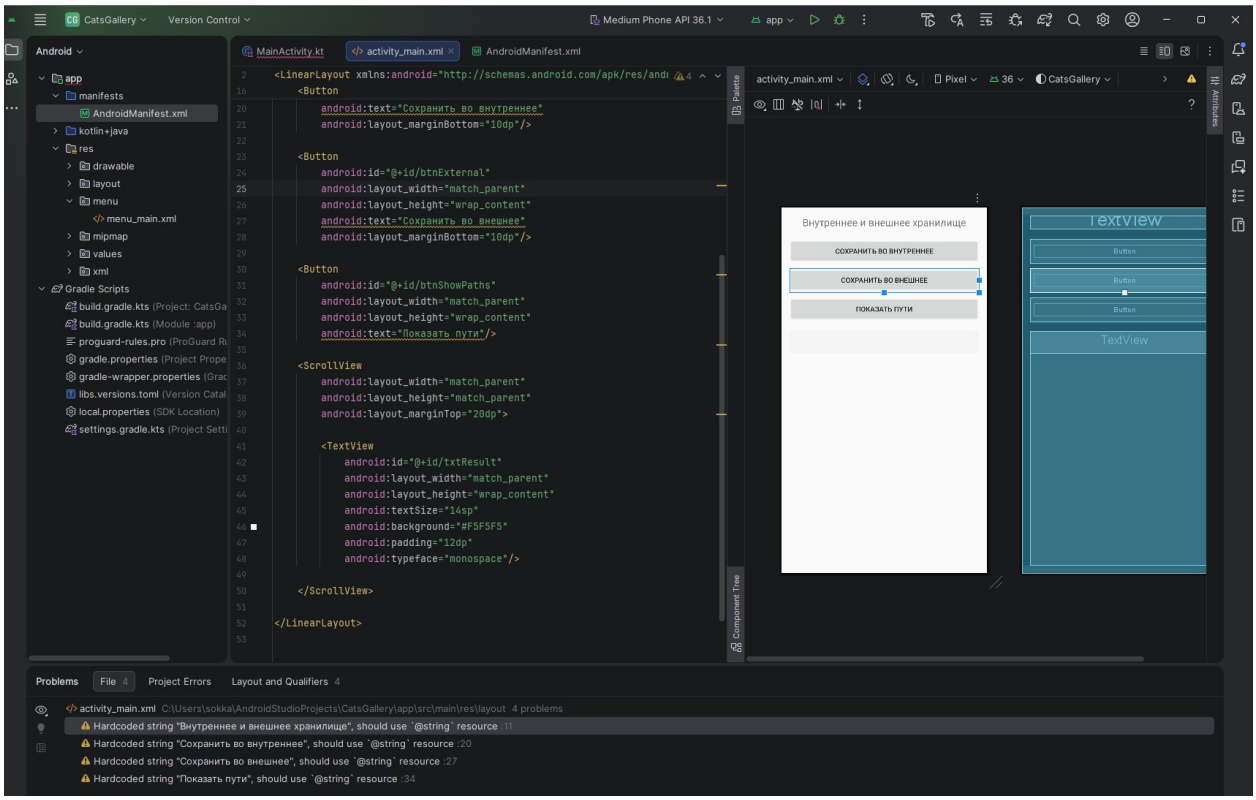
Лабораторная работа №7
Кинге В.Д.
Работа с файловой системой внутреннего и внешнего хранилища





```kotlin
package com.example.lab8storage

import android.Manifest
import android.content.pm.PackageManager
import android.os.Bundle
import android.os.Environment
import android.widget.Button
import android.widget.TextView
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity
import androidx.core.app.ActivityCompat
import androidx.core.content.ContextCompat
import java.io.File
import java.io.FileOutputStream

class MainActivity : AppCompatActivity() {

    7 Usages
    private lateinit var txtResult: TextView
    2 Usages
    private val STORAGE_PERMISSION_CODE = 100

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        txtResult = findViewById(R.id.txtResult)

        findViewById<Button>(R.id.btnInternal).setOnClickListener { saveInternal() }
        findViewById<Button>(R.id.btnExternal).setOnClickListener { saveExternal() }
        findViewById<Button>(R.id.btnShowPaths).setOnClickListener { showPaths() }
    }

    1 Usage
    private fun saveInternal() {
        try {
            val internalFile = File(filesDir, "internal_cats.txt")
```

```kotlin
class MainActivity : AppCompatActivity() {
    }

    1 Usage
    private fun saveInternal() {
        try {
            val internalFile = File(filesDir, "internal_cats.txt")
            FileOutputStream(internalFile).use { fos ->
                fos.write("Персик\nМурка\n".toByteArray())
            }
            txtResult.text = "ВНУТРЕННЕЕ ХРАНИЛИЩЕ:\n${internalFile.absolutePath}\n\nСтатус: СОХРАНЕНО"
        } catch (e: Exception) {
            txtResult.text = "ВНУТРЕННЕЕ ХРАНИЛИЩЕ:\nОшибка: ${e.message}"
        }
    }

    2 Usages
    private fun saveExternal() {
        if (ContextCompat.checkSelfPermission( context = this, permission = Manifest.permission.WRITE_EXTERNAL_STORAGE)
            != PackageManager.PERMISSION_GRANTED) {
            ActivityCompat.requestPermissions( activity = this,
                permissions = arrayOf(Manifest.permission.WRITE_EXTERNAL_STORAGE), requestCode = STORAGE_PERMISSION_CODE)
            return
        }

        if (Environment.getExternalStorageState() == Environment.MEDIA_MOUNTED) {
            try {
                val externalDir = getExternalFilesDir(Environment.DIRECTORY_DOCUMENTS)
                val externalFile = File(externalDir, "external_cats.txt")
                externalDir?.mkdirs()

                FileOutputStream(externalFile).use { fos ->
                    fos.write("Барсик\nРыжик\n".toByteArray())
                }
                txtResult.text = "ВНЕШНЕЕ ХРАНИЛИЩЕ:\n${externalFile.absolutePath}\n\nСтатус: СОХРАНЕНО"
            } catch (e: Exception) {
                txtResult.text = "ВНЕШНЕЕ ХРАНИЛИЩЕ:\nОшибка: ${e.message}"
```

```kotlin
class MainActivity : AppCompatActivity() {
    private fun saveExternal() {
        }

        if (Environment.getExternalStorageState() == Environment.MEDIA_MOUNTED) {
            try {
                val externalDir = getExternalFilesDir(Environment.DIRECTORY_DOCUMENTS)
                val externalFile = File(externalDir, "external_cats.txt")
                externalDir?.mkdirs()

                FileOutputStream(externalFile).use { fos ->
                    fos.write("Барсик\nРыжик\n".toByteArray())
                }
                txtResult.text = "ВНЕШНЕЕ ХРАНИЛИЩЕ:\n${externalFile.absolutePath}\n\nСтатус: СОХРАНЕНО"
            } catch (e: Exception) {
                txtResult.text = "ВНЕШНЕЕ ХРАНИЛИЩЕ:\nОшибка: ${e.message}"
            }
        } else {
            txtResult.text = "ВНЕШНЕЕ ХРАНИЛИЩЕ:\nSD-карта недоступна"
        }
    }

    1 Usage
    private fun showPaths() {
        val internalPath = filesDir.absolutePath
        val externalPath = getExternalFilesDir(Environment.DIRECTORY_DOCUMENTS)?.absolutePath ?: "Нет доступа"
        val publicPath = Environment.getExternalStoragePublicDirectory( type = Environment.DIRECTORY_DOWNLOADS).absolutePath

        txtResult.text = """
            ПУТИ ХРАНИЛИЩА:

            Внутреннее: $internalPath

            Внешнее (приложение): $externalPath

            Общественное: $publicPath

```

```kotlin
class MainActivity : AppCompatActivity() {
    private fun showPaths() {
            ПУТИ ХРАНИЛИЩА:

            Внутреннее: $internalPath

            Внешнее (приложение): $externalPath

            Общественное: $publicPath

            Разрешение: ${if (hasStoragePermission()) "Есть" else "Нет"}
        """.trimIndent()
    }

    1 Usage
    private fun hasStoragePermission(): Boolean {
        return ContextCompat.checkSelfPermission( context = this, permission = Manifest.permission.WRITE_EXTERNAL_STORAGE)
        == PackageManager.PERMISSION_GRANTED
    }

    override fun onRequestPermissionsResult(
        requestCode: Int,
        permissions: Array<out String>,
        grantResults: IntArray
    ) {
        super.onRequestPermissionsResult(requestCode, permissions, grantResults)
        if (requestCode == STORAGE_PERMISSION_CODE) {
            if (grantResults.isNotEmpty() && grantResults[0] == PackageManager.PERMISSION_GRANTED) {
                Toast.makeText(this, "Разрешение получено", Toast.LENGTH_SHORT).show()
                saveExternal()
            } else {
                Toast.makeText(this, "Разрешение отклонено", Toast.LENGTH_SHORT).show()
            }
        }
    }
}
```